

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

---

## Device Specifications

Device Type: Camera  
Manufacturer: ClearOne  
Firmware Version: N/A  
Model(s): UNITE 200, UNITE 100, UNITE 150

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 2.08.0002-b001                         | 1.4.2                   |

## Version History

| Module Version | Date      | Notes                                                                                                    |
|----------------|-----------|----------------------------------------------------------------------------------------------------------|
| 1_1_1_0        | 6/12/2018 | Added model UNITE 150 and RS485 support.                                                                 |
| 1_1_0_0        | 9/8/2017  | Added AutoFocus to Ethernet control. Added ImageFlip command. Changed Preset value range to '0' – '254'. |
| 1_0_1_1        | 9/5/2017  | Updated status example.                                                                                  |
| 1_0_1_0        | 8/15/2017 | Initial Version                                                                                          |

## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '1'. Device ID ranges from '1' to '7'.  
Example: `InterfaceName.DeviceID = '1'`
- UNITE 100 – Supports Serial Control Only  
UNITE 150 – Supports Serial Control Only  
UNITE 200 – Supports Both Serial and Ethernet Control Only (Ethernet control does not support any status)

## Supported Classes and Examples

|                                                                                                             |
|-------------------------------------------------------------------------------------------------------------|
| <b>SerialClass</b>                                                                                          |
| <code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='UNITE 200')</code>               |
| <b>SerialOverEthernetClass</b>                                                                              |
| <code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='UNITE 200')</code> |
| <b>EthernetClass</b>                                                                                        |
| <code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 5678, Model='UNITE 200')</code>           |

## Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                                     |                                         |                                              |                                         |
|-------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------|-----------------------------------------|
| Command<br><b>AutoExposure</b> <sup>3</sup>                                         | Value<br>'Full Auto'<br>'Iris Priority' | Value<br>'Manual'<br>'Bright'                | Value<br>'Shutter Priority'             |
| # AutoExposure example<br>InterfaceName.Set('AutoExposure', 'Full Auto')            |                                         |                                              |                                         |
| Command<br><b>AutoFocus</b>                                                         | Value<br>'Auto'                         | Value<br>'Manual'                            |                                         |
| # AutoFocus example<br>InterfaceName.Set('AutoFocus', 'Auto')                       |                                         |                                              |                                         |
| Command<br><b>Focus</b> <sup>1</sup>                                                | Value<br>'Far'                          | Value<br>'Near'                              | Value<br>'Stop'                         |
| # Focus example<br>InterfaceName.Set('Focus', 'Far')                                |                                         |                                              |                                         |
| Command<br><b>Home</b>                                                              | Value<br>None                           |                                              |                                         |
| # Home example<br>InterfaceName.Set('Home', None)                                   |                                         |                                              |                                         |
| Command<br><b>ImageFlip</b>                                                         | Value<br>'On'                           | Value<br>'Off'                               |                                         |
| Qualifier Key<br><b>'Type'</b>                                                      | Qualifier Value<br>'Horizontal'         | Qualifier Value<br>'Vertical'                |                                         |
| # ImageFlip example<br>InterfaceName.Set('ImageFlip', 'On', {'Type': 'Horizontal'}) |                                         |                                              |                                         |
| Command<br><b>Iris</b> <sup>2</sup>                                                 | Value<br>'Reset'                        | Value<br>'Up'                                | Value<br>'Down'                         |
| # Iris example<br>InterfaceName.Set('Iris', 'Reset')                                |                                         |                                              |                                         |
| Command<br><b>PanTilt</b>                                                           | Value<br>'Up'<br>'Right'<br>'Down Left' | Value<br>'Down'<br>'Up Left'<br>'Down Right' | Value<br>'Left'<br>'Up Right'<br>'Stop' |
| Qualifier Key<br><b>'Tilt Speed'</b>                                                | Qualifier Value<br>'1' – '20'           |                                              |                                         |

## Global Scripter Module Communication Sheet

|                                                                                                |                               |                           |                             |
|------------------------------------------------------------------------------------------------|-------------------------------|---------------------------|-----------------------------|
| Qualifier Key<br><b>'Pan Speed'</b>                                                            | Qualifier Value<br>'1' – '24' |                           |                             |
| # PanTilt example<br>InterfaceName.Set('PanTilt', 'Up', {'Tilt Speed': '1', 'Pan Speed': '1'}) |                               |                           |                             |
| Command<br><b>Power</b>                                                                        | Value<br>'On'                 | Value<br>'Off'            |                             |
| # Power example<br>InterfaceName.Set('Power', 'On')                                            |                               |                           |                             |
| Command<br><b>Preset</b>                                                                       | Value<br>'0' – '254'          |                           |                             |
| Qualifier Key<br><b>'Action'</b>                                                               | Qualifier Value<br>'Reset'    | Qualifier Value<br>'Save' | Qualifier Value<br>'Recall' |
| # Preset example<br>InterfaceName.Set('Preset', '0', {'Action': 'Reset'})                      |                               |                           |                             |
| Command<br><b>ResetPanTilt</b>                                                                 | Value<br>None                 |                           |                             |
| # ResetPanTilt example<br>InterfaceName.Set('ResetPanTilt', None)                              |                               |                           |                             |
| Command<br><b>Shutter</b> <sup>2</sup>                                                         | Value<br>'Reset'              | Value<br>'Up'             | Value<br>'Down'             |
| # Shutter example<br>InterfaceName.Set('Shutter', 'Reset')                                     |                               |                           |                             |
| Command<br><b>Zoom</b>                                                                         | Value<br>'Tele'               | Value<br>'Wide'           | Value<br>'Stop'             |
| Qualifier Key<br><b>'Zoom Speed'</b>                                                           | Qualifier Value<br>'0' – '7'  |                           |                             |
| # Zoom example<br>InterfaceName.Set('Zoom', 'Tele', {'Zoom Speed': '0'})                       |                               |                           |                             |

<sup>1</sup> Only works when Auto Focus is set to Manual.

<sup>2</sup> Only works when Auto Exposure is set to Manual.

<sup>3</sup> Only available for the Serial control.

## Status Available – Serial only

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                             |                                         |                               |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------|-----------------------------|
| <b>Command</b><br><b>AutoExposure</b>                                                                                                                                                       | Value<br>'Full Auto'<br>'Iris Priority' | Value<br>'Manual'<br>'Bright' | Value<br>'Shutter Priority' |
| # AutoExposure examples<br>InterfaceName.Update('AutoExposure')<br>Value = InterfaceName.ReadStatus('AutoExposure')<br>InterfaceName.SubscribeStatus('AutoExposure', None, FeedbackHandler) |                                         |                               |                             |
| <b>Command</b><br><b>AutoFocus</b>                                                                                                                                                          | Value<br>'Auto'                         | Value<br>'Manual'             |                             |
| # AutoFocus examples<br>InterfaceName.Update('AutoFocus')<br>Value = InterfaceName.ReadStatus('AutoFocus')<br>InterfaceName.SubscribeStatus('AutoFocus', None, FeedbackHandler)             |                                         |                               |                             |
| <b>Command</b><br><b>ConnectionStatus</b>                                                                                                                                                   | Value<br>'Connected'                    | Value<br>'Disconnected'       |                             |
| # ConnectionStatus examples<br>Value = InterfaceName.ReadStatus('ConnectionStatus')<br>InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)                             |                                         |                               |                             |
| <b>Command</b><br><b>Power</b>                                                                                                                                                              | Value<br>'On'                           | Value<br>'Off'                |                             |
| # Power examples<br>InterfaceName.Update('Power')<br>Value = InterfaceName.ReadStatus('Power')<br>InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)                             |                                         |                               |                             |

## Cable and Adapter Requirements

Captive Screw to Male Mini 8-Pin DIN VISCA Serial Cable.

## Notes for the Device

- RS-485 control: currently not working properly on the following models: IPCP Pro 550 and IPCP Pro 555 with firmware version 2.08.0002-b001.
- For model UNITE 150/200 RS-485 connector is shared with RS-232 Out in the camera.
- For model UNITE 100, ensure that the following settings are set in the camera menu.
  - MENU/HOME/SYSTEM OPTION/PROTOCOL: VISCA
  - MENU/HOME/SYSTEM OPTION/BAUD: 9600
  - MENU/HOME/SYSTEM OPTION/RS485: OFF (RS232 control) / ON (RS485 control)
  - MENU/HOME/SYSTEM OPTION/ADDR: 01 (Camera ID must match the driver Device ID)

## Serial communication

**Port Type:** RS-232, RS-485

**Baud Rate:** 9600

**Data Bits:** 8

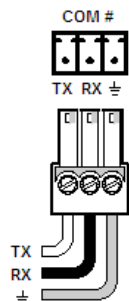
**Parity:** None

**Stop Bits:** One

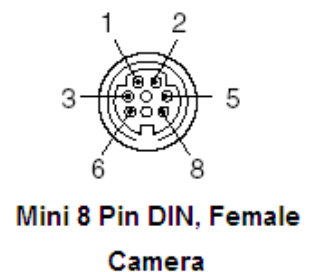
**Flow Control:** None

## Pin Assignments Diagram

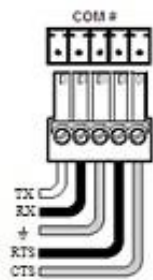
*RS232 Pin Assignments Diagram:*



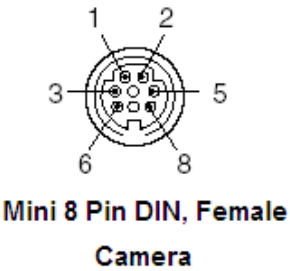
| Signal | Main Cable | Pin | Signal |
|--------|------------|-----|--------|
| TxD    | →          | 5   | RxD    |
| RxD    | ←          | 3   | TxD    |
| GND    | →          | 4   | GND    |



RS485 Pin Assignments Diagram:



| Signal | Main Cable | Pin | Signal |
|--------|------------|-----|--------|
| Tx     | →          | 6   | Data - |
| Rx     | →          | 6   | Data - |
| GND    | →          | 5   | GND    |
| RTS    | ←          | 8   | Data + |
| CTS    | ←          | 8   | Data + |



---

---

## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

|                                     |          |
|-------------------------------------|----------|
| <b>Port Type:</b>                   | Ethernet |
| <b>Default Port:</b>                | 5678     |
| <b>Logon Credentials Supported:</b> | No       |
| <b>Multi-Connection</b>             | Yes      |
| <b>Capabilities:</b>                |          |
| <b>Port Changeability:</b>          | No       |

---

---

## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device



## Appendix A. Set Commands

Commands below only work when DeviceID is set to '1'.

|                                                      |                                      |
|------------------------------------------------------|--------------------------------------|
| <b>Auto Exposure Bright</b>                          | \x81\x01\x049\x0D\xFF                |
| <b>Auto Exposure Full Auto</b>                       | \x81\x01\x049\x00\xFF                |
| <b>Auto Exposure Iris Priority</b>                   | \x81\x01\x049\x0B\xFF                |
| <b>Auto Exposure Manual</b>                          | \x81\x01\x049\x03\xFF                |
| <b>Auto Exposure Shutter Priority</b>                | \x81\x01\x049\x0A\xFF                |
| <b>Auto Focus Auto</b>                               | \x81\x01\x048\x02\xFF                |
| <b>Auto Focus Manual</b>                             | \x81\x01\x048\x03\xFF                |
| <b>Focus Far</b>                                     | \x81\x01\x04\x08\x03\xFF             |
| <b>Focus Near</b>                                    | \x81\x01\x04\x08\x02\xFF             |
| <b>Focus Stop</b>                                    | \x81\x01\x04\x08\x00\xFF             |
| <b>Home None</b>                                     | \x81\x01\x06\x04\xFF                 |
| <b>Image Flip Off Type Horizontal</b>                | \x81\x01\x04a\x03\xFF                |
| <b>Image Flip On Type Horizontal</b>                 | \x81\x01\x04a\x02\xFF                |
| <b>Image Flip Off Type Vertical</b>                  | \x81\x01\x04f\x03\xFF                |
| <b>Image Flip On Type Vertical</b>                   | \x81\x01\x04f\x02\xFF                |
| <b>Iris Down</b>                                     | \x81\x01\x04\x0B\x03\xFF             |
| <b>Iris Reset</b>                                    | \x81\x01\x04\x0B\x00\xFF             |
| <b>Iris Up</b>                                       | \x81\x01\x04\x0B\x02\xFF             |
| <b>Pan-Tilt Down Tilt Speed 1 Pan Speed 1</b>        | \x81\x01\x06\x01\x01\x01\x03\x02\xFF |
| <b>Pan-Tilt Down Left Tilt Speed 1 Pan Speed 1</b>   | \x81\x01\x06\x01\x01\x01\x01\x02\xFF |
| <b>Pan-Tilt Down Right Tilt Speed 1 Pan Speed 1</b>  | \x81\x01\x06\x01\x01\x01\x02\x02\xFF |
| <b>Pan-Tilt Left Tilt Speed 1 Pan Speed 1</b>        | \x81\x01\x06\x01\x01\x01\x01\x03\xFF |
| <b>Pan-Tilt Right Tilt Speed 1 Pan Speed 1</b>       | \x81\x01\x06\x01\x01\x01\x02\x03\xFF |
| <b>Pan-Tilt Stop Tilt Speed 1 Pan Speed 1</b>        | \x81\x01\x06\x01\x01\x01\x03\x03\xFF |
| <b>Pan-Tilt Up Tilt Speed 1 Pan Speed 1</b>          | \x81\x01\x06\x01\x01\x01\x03\x01\xFF |
| <b>Pan-Tilt Up Left Tilt Speed 1 Pan Speed 1</b>     | \x81\x01\x06\x01\x01\x01\x01\x01\xFF |
| <b>Pan-Tilt Up Right Tilt Speed 1 Pan Speed 1</b>    | \x81\x01\x06\x01\x01\x01\x02\x01\xFF |
| <b>Pan-Tilt Down Tilt Speed 1 Pan Speed 24</b>       | \x81\x01\x06\x01\x18\x01\x03\x02\xFF |
| <b>Pan-Tilt Down Left Tilt Speed 1 Pan Speed 24</b>  | \x81\x01\x06\x01\x18\x01\x01\x02\xFF |
| <b>Pan-Tilt Down Right Tilt Speed 1 Pan Speed 24</b> | \x81\x01\x06\x01\x18\x01\x02\x02\xFF |
| <b>Pan-Tilt Left Tilt Speed 1 Pan Speed 24</b>       | \x81\x01\x06\x01\x18\x01\x01\x03\xFF |
| <b>Pan-Tilt Right Tilt Speed 1 Pan Speed 24</b>      | \x81\x01\x06\x01\x18\x01\x02\x03\xFF |
| <b>Pan-Tilt Stop Tilt Speed 1 Pan Speed 24</b>       | \x81\x01\x06\x01\x18\x01\x03\x03\xFF |
| <b>Pan-Tilt Up Tilt Speed 1 Pan Speed 24</b>         | \x81\x01\x06\x01\x18\x01\x03\x01\xFF |
| <b>Pan-Tilt Up Left Tilt Speed 1 Pan Speed 24</b>    | \x81\x01\x06\x01\x18\x01\x01\x01\xFF |
| <b>Pan-Tilt Up Right Tilt Speed 1 Pan Speed 24</b>   | \x81\x01\x06\x01\x18\x01\x02\x01\xFF |

## Global Scripter Module Communication Sheet

Revision: 6/12/2018

|                                                       |                                      |
|-------------------------------------------------------|--------------------------------------|
| <b>Pan-Tilt Down Tilt Speed 20 Pan Speed 1</b>        | \x81\x01\x06\x01\x01\x14\x03\x02\xff |
| <b>Pan-Tilt Down Left Tilt Speed 20 Pan Speed 1</b>   | \x81\x01\x06\x01\x01\x14\x01\x02\xff |
| <b>Pan-Tilt Down Right Tilt Speed 20 Pan Speed 1</b>  | \x81\x01\x06\x01\x01\x14\x02\x02\xff |
| <b>Pan-Tilt Left Tilt Speed 20 Pan Speed 1</b>        | \x81\x01\x06\x01\x01\x14\x01\x03\xff |
| <b>Pan-Tilt Right Tilt Speed 20 Pan Speed 1</b>       | \x81\x01\x06\x01\x01\x14\x02\x03\xff |
| <b>Pan-Tilt Stop Tilt Speed 20 Pan Speed 1</b>        | \x81\x01\x06\x01\x01\x14\x03\x03\xff |
| <b>Pan-Tilt Up Tilt Speed 20 Pan Speed 1</b>          | \x81\x01\x06\x01\x01\x14\x03\x01\xff |
| <b>Pan-Tilt Up Left Tilt Speed 20 Pan Speed 1</b>     | \x81\x01\x06\x01\x01\x14\x01\x01\xff |
| <b>Pan-Tilt Up Right Tilt Speed 20 Pan Speed 1</b>    | \x81\x01\x06\x01\x01\x14\x02\x01\xff |
| <b>Pan-Tilt Down Tilt Speed 20 Pan Speed 24</b>       | \x81\x01\x06\x01\x18\x14\x03\x02\xff |
| <b>Pan-Tilt Down Left Tilt Speed 20 Pan Speed 24</b>  | \x81\x01\x06\x01\x18\x14\x01\x02\xff |
| <b>Pan-Tilt Down Right Tilt Speed 20 Pan Speed 24</b> | \x81\x01\x06\x01\x18\x14\x02\x02\xff |
| <b>Pan-Tilt Left Tilt Speed 20 Pan Speed 24</b>       | \x81\x01\x06\x01\x18\x14\x01\x03\xff |
| <b>Pan-Tilt Right Tilt Speed 20 Pan Speed 24</b>      | \x81\x01\x06\x01\x18\x14\x02\x03\xff |
| <b>Pan-Tilt Stop Tilt Speed 20 Pan Speed 24</b>       | \x81\x01\x06\x01\x18\x14\x03\x03\xff |
| <b>Pan-Tilt Up Tilt Speed 20 Pan Speed 24</b>         | \x81\x01\x06\x01\x18\x14\x03\x01\xff |
| <b>Pan-Tilt Up Left Tilt Speed 20 Pan Speed 24</b>    | \x81\x01\x06\x01\x18\x14\x01\x01\xff |
| <b>Pan-Tilt Up Right Tilt Speed 20 Pan Speed 24</b>   | \x81\x01\x06\x01\x18\x14\x02\x01\xff |
| <b>Power Off</b>                                      | \x81\x01\x04\x00\x03\xff             |
| <b>Power On</b>                                       | \x81\x01\x04\x00\x02\xff             |
| <b>Preset 0 Action Recall</b>                         | \x81\x01\x04?\x02\x00\xff            |
| <b>Preset 254 Action Recall</b>                       | \x81\x01\x04?\x02\xFE\xff            |
| <b>Preset 0 Action Reset</b>                          | \x81\x01\x04?\x00\x00\xff            |
| <b>Preset 254 Action Reset</b>                        | \x81\x01\x04?\x00\xFE\xff            |
| <b>Preset 0 Action Save</b>                           | \x81\x01\x04?\x01\x00\xff            |
| <b>Preset 254 Action Save</b>                         | \x81\x01\x04?\x01\xFE\xff            |
| <b>Reset Pan-Tilt None</b>                            | \x81\x01\x06\x05\xff                 |
| <b>Shutter Down</b>                                   | \x81\x01\x04\x0A\x03\xff             |
| <b>Shutter Reset</b>                                  | \x81\x01\x04\x0A\x00\xff             |
| <b>Shutter Up</b>                                     | \x81\x01\x04\x0A\x02\xff             |
| <b>Zoom Stop Zoom Speed 0</b>                         | \x81\x01\x04\x07\x00\xff             |
| <b>Zoom Tele Zoom Speed 0</b>                         | \x81\x01\x04\x07 \xff                |
| <b>Zoom Wide Zoom Speed 0</b>                         | \x81\x01\x04\x070\xff                |
| <b>Zoom Stop Zoom Speed 7</b>                         | \x81\x01\x04\x07\x00\xff             |
| <b>Zoom Tele Zoom Speed 7</b>                         | \x81\x01\x04\x07'\xff                |
| <b>Zoom Wide Zoom Speed 7</b>                         | \x81\x01\x04\x077\xff                |

---

## Appendix B. Update Commands

Commands below only work when DeviceID is set to '1'.

|                      |                      |
|----------------------|----------------------|
| <b>Auto Exposure</b> | \x81\x09\x049\xff    |
| <b>Auto Focus</b>    | \x81\x09\x048\xff    |
| <b>Power</b>         | \x81\x09\x04\x00\xff |
| <b>Zoom Position</b> | \x81\x09\x04G\xff    |